

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks: 50

**Annexure A
Scenario-Based**

Code of Conduct:

I **Jannathul Firdaus N (192573025)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

- School of Computing → /25
- School of Electronics → /26
- School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

- 1. Determine the network address and broadcast address for each department.
- 2. Identify the valid host IP address range for each subnet.
- 3. Calculate the number of usable host addresses available in each subnet.
- 4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

- Software Development → /24
- Quality Assurance → /25
- Human Resources → /26
- Executive Management → /27

As the network administrator, perform the following tasks:

- 1. Determine the network address and broadcast address for each department.

- 2. Identify the valid range of host IP addresses.
- 3. Calculate the number of usable hosts in each subnet.
- 4. Identify the unused IP address range remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

- 1. Convert the given IPv6 address into its compressed notation.
- 2. Expand the compressed address back into its full hexadecimal representation.
- 3. Identify the network prefix using the /64 prefix length.
- 4. Calculate the total number of host addresses supported within the subnet.
- 5. The total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

A packet arrives at the router with the destination IP address:
192.168.2.45

As the routing engineer, determine:

- 1. The destination subnet to which the packet belongs.
- 2. The matching routing table entry.
- 3. The next-hop network/interface used to forward the packet.

- 4. The default route that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

- Administration LAN → 192.168.10.0/26
- Finance LAN → 192.168.10.64/26

The router interfaces are configured with:

- Administration Gateway → 192.168.10.1
- Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

- 1. Determine the network address and broadcast address for each LAN.
- 2. Identify the valid host IP address range.
- 3. Suggest suitable IP addresses for three computers in each LAN.
- 4. Identify the default gateway that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

ANSWERS TO SCENARIO-BASED QUESTIONS

Question 1: Smart Campus Subnet Allocation (192.168.100.0/24)

The block 192.168.100.0/24 (256 addresses) is divided using VLSM, allocating the largest subnet first (/25), followed by /26 and /27.

1 & 2. Network Address, Broadcast Address and Valid Host Range

Department	Subnet Mask	Network Address	Broadcast Address	Valid Host Range
School of Computing	/25 (255.255.255.128)	192.168.100.0	192.168.100.127	192.168.100.1 - 192.168.100.126
School of Electronics	/26 (255.255.255.192)	192.168.100.128	192.168.100.191	192.168.100.129 - 192.168.100.190
School of Mechanical Engg.	/27 (255.255.255.224)	192.168.100.192	192.168.100.223	192.168.100.193 - 192.168.100.222

3. Usable Host Addresses per Subnet

- School of Computing (/25): $2^7 - 2 = 126$ usable hosts
- School of Electronics (/26): $2^6 - 2 = 62$ usable hosts
- School of Mechanical Engineering (/27): $2^5 - 2 = 30$ usable hosts

4. Remaining Unused IP Address Range

After allocating the three subnets (192.168.100.0 - 192.168.100.223), the remaining unused block is:

192.168.100.224 - 192.168.100.255 (a /27-sized block of 32 addresses, reserved for future departments).

Question 2: Multinational Company Subnet Allocation (10.10.0.0/16)

The block 10.10.0.0/16 is divided using VLSM. Subnets are allocated sequentially in descending order of size: /24, /25, /26, /27.

1 & 2. Network Address, Broadcast Address and Valid Host Range

Department	Subnet Mask	Network Address	Broadcast Address	Valid Host Range
Software Development	/24 (255.255.255.0)	10.10.0.0	10.10.0.255	10.10.0.1 - 10.10.0.254
Quality Assurance	/25 (255.255.255.128)	10.10.1.0	10.10.1.127	10.10.1.1 - 10.10.1.126
Human Resources	/26 (255.255.255.192)	10.10.1.128	10.10.1.191	10.10.1.129 - 10.10.1.190
Executive Management	/27 (255.255.255.224)	10.10.1.192	10.10.1.223	10.10.1.193 - 10.10.1.222

3. Usable Hosts per Subnet

- Software Development (/24): $2^8 - 2 = 254$ usable hosts
- Quality Assurance (/25): $2^7 - 2 = 126$ usable hosts
- Human Resources (/26): $2^6 - 2 = 62$ usable hosts
- Executive Management (/27): $2^5 - 2 = 30$ usable hosts

4. Unused IP Address Range Remaining

After allocating the four departments (10.10.0.0 - 10.10.1.223), the remaining unused range within the /16 block is:

10.10.1.224 - 10.10.255.255 (a very large block available for future expansion).

Question 3: IPv6 Addressing for Smart Hospital (2001:0db8:abcd::/64)

1. Compressed Notation

2001:0db8:abcd:0000:0000:0000:0000:0000 compresses to:
2001:db8:abcd::

(Leading zeros in each group are dropped, and the longest run of consecutive all-zero groups is replaced with '::'.)

2. Expanded (Full Hexadecimal) Representation

2001:db8:abcd:: expands back to:
2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network Prefix (/64)

The first 64 bits form the network prefix: 2001:0db8:abcd:0000 (i.e., 2001:0db8:abcd:0000::/64). The remaining 64 bits are available for the interface/host identifier.

4. Total Host Addresses Supported in the Subnet

With a /64 prefix, $128 - 64 = 64$ bits remain for host addressing.
Total host addresses = $2^{64} = 18,446,744,073,709,551,616$ addresses.

5. Total Possible Host Addresses Available in this Network

Since IPv6 subnets do not reserve a separate network/broadcast address the way IPv4 does, all 2^{64} (18,446,744,073,709,551,616) addresses are usable for hosts/IoT devices within this /64 subnet.

Question 4: Routing Table Lookup at ISP Core Router

Routing table entries: 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24.

Destination IP of arriving packet: 192.168.2.45

1. Destination Subnet

192.168.2.45 falls within the range 192.168.2.0 - 192.168.2.255, so it belongs to the subnet 192.168.2.0/24.

2. Matching Routing Table Entry

The router performs a longest-prefix match and finds that 192.168.2.0/24 matches the destination address.

3. Next-Hop Network/Interface

The packet is forwarded out of the router interface directly connected to the 192.168.2.0/24 network (the interface configured with an IP in that subnet, e.g., the LAN interface serving Customer Network 2).

4. Default Route (if no match found)

If the destination address did not match any entry in the routing table, the router would forward the packet using the default route 0.0.0.0/0, typically directed toward the ISP's upstream gateway/next-hop router.

Question 5: Two-Department LAN via Layer-3 Router

1 & 2. Network Address, Broadcast Address and Valid Host Range

LAN	Subnet	Network Address	Broadcast Address	Valid Host Range
Administration LAN	192.168.10.0/26	192.168.10.0	192.168.10.63	192.168.10.1 - 192.168.10.62
Finance LAN	192.168.10.64/26	192.168.10.64	192.168.10.127	192.168.10.65 - 192.168.10.126

3. Suggested IP Addresses for Three Computers in Each LAN

- Administration LAN: 192.168.10.2, 192.168.10.3, 192.168.10.4 (Note: 192.168.10.1 is reserved for the gateway)
- Finance LAN: 192.168.10.66, 192.168.10.67, 192.168.10.68 (Note: 192.168.10.65 is reserved for the gateway)

4. Default Gateway for Each LAN

- Administration LAN hosts use default gateway: 192.168.10.1
- Finance LAN hosts use default gateway: 192.168.10.65